Linear Algebra: Tutorial Session 2

Problem 1

Prove that the following two axioms:

- $\exists \mathbf{0} \in V$ such that $\boldsymbol{\alpha} + \mathbf{0} = \mathbf{0} + \boldsymbol{\alpha} = \boldsymbol{\alpha}$.
- $\forall \boldsymbol{\alpha} \in V$, $\exists \boldsymbol{\beta} \in V$ such that $\boldsymbol{\alpha} + \boldsymbol{\beta} = \boldsymbol{\beta} + \boldsymbol{\alpha} = \mathbf{0}$.

is equivalent with the following statement:

- $\forall \boldsymbol{\alpha}, \boldsymbol{\beta} \in V$, $\exists \boldsymbol{x} \in V$ such that $\boldsymbol{\alpha} + \boldsymbol{x} = \boldsymbol{\beta}$.

When trying to prove the equivalence, you may use the rest of the six axioms of vector space. Strictly speaking, we are proving the equivalence conditioned on the other six axioms.

Answer:

Let (A) denote the axiom of the zero vector, (B) the axiom of additive inverses, and (C) the statement in the question. We work over the other six axioms: the commutativity and associativity of addition, and the four axioms on scalar multiplication. We also use that V is a nonempty set, as in the definition of a vector space.

$(A) \wedge (B) \implies (C)$. Given $\boldsymbol{\alpha}, \boldsymbol{\beta} \in V$, axiom (B) gives an $\boldsymbol{\alpha}' \in V$ with $\boldsymbol{\alpha} + \boldsymbol{\alpha}' = \mathbf{0}$. Take $\boldsymbol{x} = \boldsymbol{\alpha}' + \boldsymbol{\beta}$. Then

$$\boldsymbol{\alpha} + \boldsymbol{x} = \boldsymbol{\alpha} + (\boldsymbol{\alpha}' + \boldsymbol{\beta}) = (\boldsymbol{\alpha} + \boldsymbol{\alpha}') + \boldsymbol{\beta} = \mathbf{0} + \boldsymbol{\beta} = \boldsymbol{\beta},$$

where the second equality is associativity, the third uses (B) and the last one is axiom (A).

$(C) \implies (A) \wedge (B)$. Since V is nonempty, fix an $\boldsymbol{a} \in V$. Applying (C) with $\boldsymbol{\alpha} = \boldsymbol{\beta} = \boldsymbol{a}$ gives an $\boldsymbol{e} \in V$ with $\boldsymbol{a} + \boldsymbol{e} = \boldsymbol{a}$. We claim that $\boldsymbol{e}$ is a zero vector for all of V. For an arbitrary $\boldsymbol{b} \in V$, applying (C) with $\boldsymbol{\alpha} = \boldsymbol{a}$ and $\boldsymbol{\beta} = \boldsymbol{b}$ gives a $\boldsymbol{y} \in V$ with $\boldsymbol{a} + \boldsymbol{y} = \boldsymbol{b}$, and hence

$$\boldsymbol{b} + \boldsymbol{e} = (\boldsymbol{a} + \boldsymbol{y}) + \boldsymbol{e} = \boldsymbol{a} + (\boldsymbol{y} + \boldsymbol{e}) = \boldsymbol{a} + (\boldsymbol{e} + \boldsymbol{y}) = (\boldsymbol{a} + \boldsymbol{e}) + \boldsymbol{y} = \boldsymbol{a} + \boldsymbol{y} = \boldsymbol{b},$$

using associativity and commutativity of addition. By commutativity, $\boldsymbol{e} + \boldsymbol{b} = \boldsymbol{b}$ as well, so $\boldsymbol{e}$ satisfies (A) with $\mathbf{0} = \boldsymbol{e}$. Finally, for an arbitrary $\boldsymbol{\alpha} \in V$, applying (C) with $\boldsymbol{\beta} = \mathbf{0}$ gives a $\boldsymbol{\beta}' \in V$ with $\boldsymbol{\alpha} + \boldsymbol{\beta}' = \mathbf{0}$, and $\boldsymbol{\beta}' + \boldsymbol{\alpha} = \mathbf{0}$ by commutativity. This is (B), so the two axioms are equivalent to (C).

Problem 2

Consider the following subset V of $\mathbb{R}^{3\times 3}$: The elements of V are of the form

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix},$$

such that the sum of the elements in each row, column, the main diagonal, and the anti-diagonal are equal. For example, the following matrix is an element of V:

$$\begin{bmatrix} 6 & 1 & 8 \\ 7 & 5 & 3 \\ 2 & 9 & 4 \end{bmatrix}.$$

1. Prove that V is a linear subspace of $\mathbb{R}^{3\times 3}$.

2. Determine the dimension and a basis of V.

Hint: Can you find a linear transformation such that V is the null space?

Answer:

For a matrix $A \in \mathbb{R}^{3\times 3}$ let r_1, r_2, r_3 be its row sums, c_1, c_2, c_3 its column sums, d its main diagonal sum and g its anti-diagonal sum; these are the eight sums of A.

1. Consider the linear transformation

$$T : \mathbb{R}^{3\times 3} \longrightarrow \mathbb{R}^7, \qquad T(A) = (r_1 - r_2,\ r_2 - r_3,\ c_1 - c_2,\ c_2 - c_3,\ r_1 - c_1,\ d - r_1,\ g - r_1).$$

The eight sums of A are equal if and only if these seven differences all vanish, so

$$V = \mathcal{N}(T),$$

the null space of a linear transformation, which is a linear subspace of $\mathbb{R}^{3\times 3}$.

2. Identify $\mathbb{R}^{3\times 3}$ with $\mathbb{R}^9$ by writing a matrix out in the order

$$(a_{11}, a_{12}, a_{13}, a_{21}, a_{22}, a_{23}, a_{31}, a_{32}, a_{33}),$$

and take this basis together with the standard basis of $\mathbb{R}^7$. Each of the seven components of T is linear in the nine entries, so T is represented by a 7×9 matrix, whose rows we read off from the definition; for example

$$r_1 - r_2 = (a_{11} + a_{12} + a_{13}) - (a_{21} + a_{22} + a_{23})$$

gives the first row $(1, 1, 1, -1, -1, -1, 0, 0, 0)$, and

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$$[T] = \begin{bmatrix} 1 & 1 & 1 & -1 & -1 & -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 & 1 & -1 & -1 & -1 \\ 1 & -1 & 0 & 1 & -1 & 0 & 1 & -1 & 0 \\ 0 & 1 & -1 & 0 & 1 & -1 & 0 & 1 & -1 \\ 0 & 1 & 1 & -1 & 0 & 0 & -1 & 0 & 0 \\ 0 & -1 & -1 & 0 & 1 & 0 & 0 & 0 & 1 \\ -1 & -1 & 0 & 0 & 1 & 0 & 1 & 0 & 0 \end{bmatrix}.$$

Row reduction gives

$$\begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 & -2/3 & -2/3 & 1/3 \\ 0 & 1 & 0 & 0 & 0 & 0 & -2/3 & 1/3 & -2/3 \\ 0 & 0 & 1 & 0 & 0 & 0 & 1/3 & -2/3 & -2/3 \\ 0 & 0 & 0 & 1 & 0 & 0 & 2/3 & -1/3 & -4/3 \\ 0 & 0 & 0 & 0 & 1 & 0 & -1/3 & -1/3 & -1/3 \\ 0 & 0 & 0 & 0 & 0 & 1 & -4/3 & -1/3 & 2/3 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix},$$

which has six nonzero rows, so $\operatorname{rank}(T) = 6$: the seven components of T are subject to exactly one linear relation. Since $\dim \mathbb{R}^{3\times 3} = 9$,

$$\dim V = \operatorname{nullity}(T) = \dim \mathbb{R}^{3\times 3} - \operatorname{rank}(T) = 9 - 6 = 3.$$

Since $V = \mathcal{N}(T)$, a basis is obtained by solving $[T]\boldsymbol{x} = \mathbf{0}$; taking x_7, x_8, x_9 as the free variables and clearing denominators gives the three matrices

$$B_1 = \begin{bmatrix} 2 & 2 & -1 \\ -2 & 1 & 4 \\ 3 & 0 & 0 \end{bmatrix}, \quad B_2 = \begin{bmatrix} 2 & -1 & 2 \\ 1 & 1 & 1 \\ 0 & 3 & 0 \end{bmatrix}, \quad B_3 = \begin{bmatrix} -1 & 2 & 2 \\ 4 & 1 & -2 \\ 0 & 0 & 3 \end{bmatrix},$$

which span V and are linearly independent, so they form a basis of V. More compactly, since $B_1 + B_2 + B_3 = 3J$, the same space is spanned by

$$J = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}, \quad M_1 = \begin{bmatrix} 1 & -1 & 0 \\ -1 & 0 & 1 \\ 0 & 1 & -1 \end{bmatrix}, \quad M_2 = \begin{bmatrix} 0 & 1 & -1 \\ -1 & 0 & 1 \\ 1 & -1 & 0 \end{bmatrix}.$$

For J all eight sums equal 3, and for M_1 and M_2 all eight sums equal 0, so all three matrices lie in $V = \mathcal{N}(T)$. They are linearly independent: the $(2,2)$-entry of $\lambda_1 M_1 + \lambda_2 M_2 + \mu J$ is μ, so $\mu = 0$, and then the $(1,1)$-entry $\lambda_1 = 0$ and the $(1,3)$-entry $-\lambda_2 = 0$ give $\lambda_1 = \lambda_2 = 0$. Three linearly independent vectors in the 3-dimensional space V form a basis of V.

Problem 3

Let V be a vector space of finite dimension n, and W_1, W_2 be two subspaces of V with dimension no larger than $m < n$. Prove that there exists a linear subspace W of V that is of dimension $n - m$, and $W \cap W_1 = W \cap W_2 = \{\mathbf{0}\}$.

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Hint: Recall from homework we proved that $W_1 \cup W_2$ is a subspace of V if and only if $W_1 \subseteq W_2$ or $W_2 \subseteq W_1$. What does this tell you for the case $m = n - 1$?

Answer:

Put $t = n - m$, so $t \geq 1$. We prove the claim by induction on t, and we shall use the following fact. Since $\dim W_1, \dim W_2 \leq m < n$, both W_i are proper subspaces of V; if $W_1 \cup W_2$ were a subspace, then by the homework result $W_1 \subseteq W_2$ or $W_2 \subseteq W_1$, and in either case $W_1 \cup W_2$ equals the larger of the two, which is proper. Hence

$$W_1 \cup W_2 \neq V,$$

so a vector $\boldsymbol{v} \notin W_1 \cup W_2$ always exists.

$t = 1$, that is $m = n - 1$. Take $\boldsymbol{v} \notin W_1 \cup W_2$ and put $W = \mathrm{Span}\{\boldsymbol{v}\}$. Then $\dim W = 1 = n - m$, and $W \cap W_i = \{\mathbf{0}\}$ for $i = 1, 2$, because every nonzero vector of W is a nonzero multiple of $\boldsymbol{v}$, which lies outside W_i.

$t \geq 2$, that is $m \leq n - 2$. Take $\boldsymbol{v} \notin W_1 \cup W_2$ and consider the quotient space $\bar{V} = V/\mathrm{Span}\{\boldsymbol{v}\}$ with the quotient map $\pi : V \to \bar{V}$, so $\dim \bar{V} = n - 1$ and $m \leq n - 2 < n - 1$. The subspaces $\bar{W}_i = \pi(W_i)$ satisfy $\dim \bar{W}_i \leq \dim W_i \leq m$, so the induction hypothesis, applied in the $(n-1)$-dimensional space $\bar{V}$ with the same m, gives a subspace $\bar{W}$ of $\bar{V}$ with

$$\dim \bar{W} = (n - 1) - m$$

and $\bar{W} \cap \bar{W}_1 = \bar{W} \cap \bar{W}_2 = \{\bar{\mathbf{0}}\}$. Put $W = \pi^{-1}(\bar{W})$. Since π is linear and its null space is $\mathrm{Span}\{\boldsymbol{v}\}$,

$$\dim W = \dim \bar{W} + \dim \mathrm{Span}\{\boldsymbol{v}\} = (n - 1 - m) + 1 = n - m.$$

Finally, take $\boldsymbol{x} \in W \cap W_1$. Then $\pi(\boldsymbol{x}) \in \bar{W} \cap \bar{W}_1 = \{\bar{\mathbf{0}}\}$, so $\boldsymbol{x} \in \mathrm{Span}\{\boldsymbol{v}\}$; but $\boldsymbol{x} \in W_1$ as well, and $\mathrm{Span}\{\boldsymbol{v}\} \cap W_1 = \{\mathbf{0}\}$ by the choice of $\boldsymbol{v}$. Hence $\boldsymbol{x} = \mathbf{0}$. The same argument with W_2 gives $W \cap W_2 = \{\mathbf{0}\}$, and the induction is complete.

Problem 4

Given vector space V. We call a linear transformation $T : V \to V$ idempotent if $T^2 = T$. Prove that if $\sigma, \tau, \sigma+\tau : V \to V$ are idempotent linear transformations from V to V, then $\sigma\tau = 0$ is the zero transformation.

Answer:

Since σ, τ and $\sigma + \tau$ are idempotent, we have $\sigma^2 = \sigma$, $\tau^2 = \tau$ and

$$(\sigma + \tau)^2 = \sigma + \tau.$$

Expanding the left-hand side by linearity gives $\sigma^2 + \sigma\tau + \tau\sigma + \tau^2 = \sigma + \tau$, that is,

$$\sigma\tau + \tau\sigma = 0,$$

where 0 denotes the zero transformation.

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Composing this identity with σ on the left gives $\sigma\tau + \sigma\tau\sigma = 0$, and composing it with σ on the right gives $\sigma\tau\sigma + \tau\sigma = 0$. Subtracting the second identity from the first yields $\sigma\tau - \tau\sigma = 0$, so $\sigma\tau = \tau\sigma$. Substituting into $\sigma\tau + \tau\sigma = 0$ gives $2\sigma\tau = 0$. Since $2 \neq 0$ in the field of scalars, $\sigma\tau = 0$.

Problem 5

Assume V_1, V_2, V_3 are three vector spaces defined on the same field $\mathbb{F}$. Let $\varphi : V_1 \to V_2$, $\psi : V_1 \to V_3$ be two linear transformations. Prove that the following statements are equivalent:

- There exists linear transformation $\sigma : V_2 \to V_3$ such that $\psi = \sigma \circ \varphi$.
- The null space of φ is a subspace of the null space of ψ.

A further question: Can you identify a necessary and sufficient condition about φ, ψ and the vector spaces V_1, V_2, V_3 such that not only σ exists, but also σ is unique?

Answer:

(i) $\implies$ (ii). Let $\boldsymbol{x} \in \mathcal{N}(\varphi)$. Then $\psi(\boldsymbol{x}) = \sigma(\varphi(\boldsymbol{x})) = \sigma(\boldsymbol{0}) = \boldsymbol{0}$, so $\boldsymbol{x} \in \mathcal{N}(\psi)$.

(ii) $\implies$ (i). Assume $\mathcal{N}(\varphi) \subseteq \mathcal{N}(\psi)$. For $\boldsymbol{y} \in \mathcal{R}(\varphi)$ choose $\boldsymbol{x} \in V_1$ with $\varphi(\boldsymbol{x}) = \boldsymbol{y}$ and set $\sigma_0(\boldsymbol{y}) = \psi(\boldsymbol{x})$. This does not depend on the choice of $\boldsymbol{x}$: if $\varphi(\boldsymbol{x}) = \varphi(\boldsymbol{x}')$, then $\boldsymbol{x} - \boldsymbol{x}' \in \mathcal{N}(\varphi) \subseteq \mathcal{N}(\psi)$, hence $\psi(\boldsymbol{x}) = \psi(\boldsymbol{x}')$. The map $\sigma_0 : \mathcal{R}(\varphi) \to V_3$ is linear, because $\varphi(\boldsymbol{x} + \boldsymbol{x}') = \boldsymbol{y} + \boldsymbol{y}'$ gives

$$\sigma_0(\boldsymbol{y} + \boldsymbol{y}') = \psi(\boldsymbol{x} + \boldsymbol{x}') = \psi(\boldsymbol{x}) + \psi(\boldsymbol{x}') = \sigma_0(\boldsymbol{y}) + \sigma_0(\boldsymbol{y}'),$$

and $\varphi(c\boldsymbol{x}) = c\boldsymbol{y}$ gives $\sigma_0(c\boldsymbol{y}) = \psi(c\boldsymbol{x}) = c\psi(\boldsymbol{x}) = c\sigma_0(\boldsymbol{y})$.

It remains to extend σ_0 to all of V_2. Choose a subspace C of V_2 with

$$V_2 = \mathcal{R}(\varphi) \oplus C,$$

for instance by extending a basis of $\mathcal{R}(\varphi)$ to a basis of V_2 and letting C be the span of the added vectors. Every $\boldsymbol{v} \in V_2$ is then uniquely $\boldsymbol{v} = \varphi(\boldsymbol{x}) + \boldsymbol{c}$ with $\boldsymbol{c} \in C$; define $\sigma(\boldsymbol{v}) = \sigma_0(\varphi(\boldsymbol{x})) = \psi(\boldsymbol{x})$. This is a linear map $V_2 \to V_3$, since it is linear on each summand of a direct sum, and for $\boldsymbol{x} \in V_1$ we get $\sigma(\varphi(\boldsymbol{x})) = \psi(\boldsymbol{x})$, that is, $\psi = \sigma \circ \varphi$.

As for the further question, the construction shows that every solution σ is determined on $\mathcal{R}(\varphi)$ by $\sigma(\varphi(\boldsymbol{x})) = \psi(\boldsymbol{x})$, while on a complement C with $V_2 = \mathcal{R}(\varphi) \oplus C$ it may be an arbitrary linear map $C \to V_3$, and different choices give different solutions. Hence

$$\sigma \text{ is unique} \iff \text{the only linear map } C \to V_3 \text{ is the zero map} \iff C = \{\boldsymbol{0}\} \text{ or } V_3 = \{\boldsymbol{0}\}.$$

Since $C = \{\boldsymbol{0}\}$ means $\mathcal{R}(\varphi) = V_2$, that is, φ is onto, the answer is: assuming a solution exists, which by (ii) means $\mathcal{N}(\varphi) \subseteq \mathcal{N}(\psi)$, the map σ is unique if and only if φ is surjective, apart from the degenerate case $V_3 = \{\boldsymbol{0}\}$, in which $\sigma = 0$ is the only map in any case.